Mid-term Examination

PHYSICS · Paper 4 (A2)

Centre No.: _____ **Class:** _____

Candidate No.: _____ **Candidate name:** _____

Exam Duration: 90 mins

Additional Materials: Answer Booklet/Paper

Graph Paper · List of Formulae

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, Candidate number and name on all the work you hand in.

Write in dark blue or black pen. You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer all questions.

Give non-exact numerical answers correct to 3 significant figures, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 100.

Topics: Motion in a circle · Gravitational fields · Oscillations

5 questions · 100 marks

Question 1 · Motion in a circle · 20 marks

Circular motion of a clock's minute hand

1 **(a)** Define the radian.

.....

..... [2]

(b) The minute hand of a clock revolves at constant angular speed around the face of the clock, completing one revolution every hour. A small piece of modelling clay is attached to the hand with its centre of gravity at a distance *L* from the fixed end of the hand, as shown in Fig. 1.1.

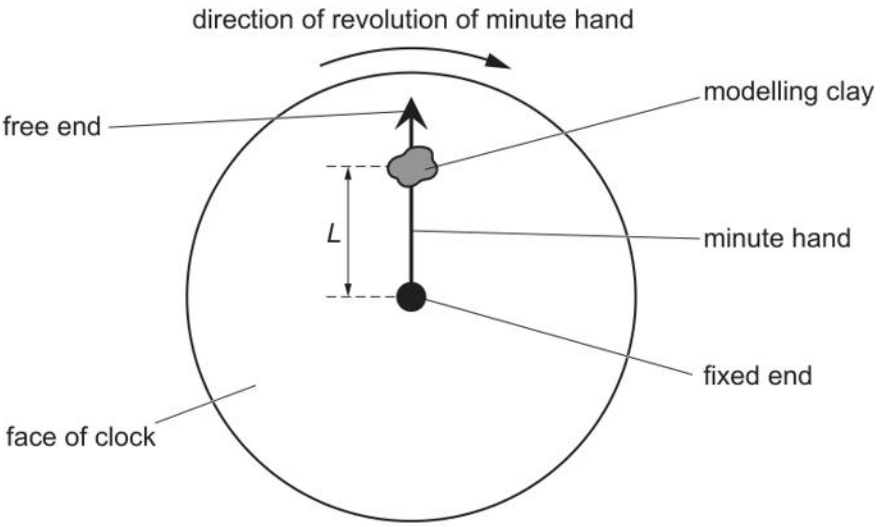


Fig. 1.1

Calculate the angular speed ω of the minute hand.

ω = ... rad s^{-1} [4]

(c) During a time interval of 1400 s, the centre of gravity of the piece of modelling clay in Fig. 1.1 moves through a total distance of 0.44 m.

(i) Calculate the angle through which the minute hand moves in this time interval.

angle = ... rad [2]

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(ii) Determine distance L.

L = ... m [4]

(iii) Calculate the magnitude of the centripetal acceleration of the piece of modelling clay.

centripetal acceleration = ... $m\,s^{-2}$ [4]

(d) Use your answer in **(c)(iii)** to explain why the variation with time of the magnitude of the force exerted by the minute hand on the piece of modelling clay is negligible as the minute hand undergoes one full revolution.

.....

.....

..... [4]

[Total: 20]

Question 2 · Gravitational fields · 20 marks

Gravitational potential of a planet — surface g & satellite orbit period

1 **(a)** Define gravitational potential at a point.

.....

.....

..... [3]

(b) Artemis is a spherical planet that may be assumed to be isolated in space. The variation with distance x from the centre of Artemis of the gravitational potential ϕ is shown in Fig. 1.1.

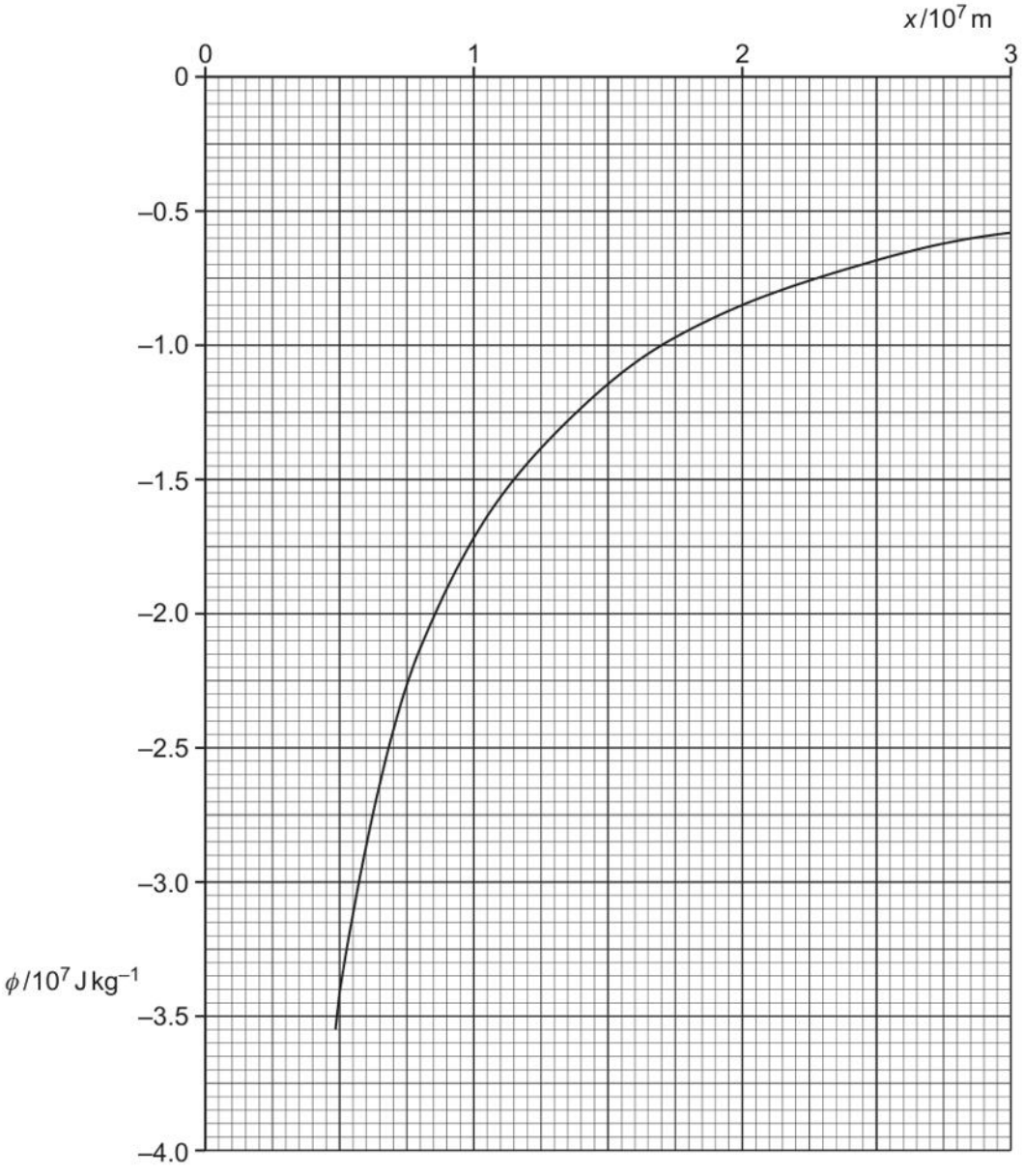


Fig. 1.1

(i) The radius of Artemis is 4800 km.

Determine the value of ϕ on the surface of Artemis.

ϕ = ... $J\,kg^{-1}$ [2]

(ii) Show that the mass of Artemis is 2.55×10^{24} kg.

[2]

(iii) Calculate the gravitational field strength g on the surface of Artemis.

g = ... $N\,kg^{-1}$ [3]

(iv) A satellite is in an orbit at a fixed position above a point on the surface of Artemis. The satellite is located above the equator of Artemis at a height above the surface where the gravitational potential is $-0.65 \times 10^{7}\,J\,kg^{-1}$.

Calculate the period, in hours, of rotation of Artemis.

period = ... hours [7]

(c) State **one** similarity and **one** difference between gravitational potential due to a point mass and electric potential due to a point charge.

similarity .....

.....

difference .....

.....

[3]

[Total: 20]

Question 3 · Gravitational fields · 20 marks

Gravitational potential & escape speed — the Moon's field & hydrogen atmosphere

2 **(a)** **(i)** Define gravitational potential at a point.

.....

.....

..... [3]

(ii) The Moon may be considered to be an isolated uniform sphere of mass 7.3×10^{22} kg and radius 1.7×10^{6} m.

Calculate the gravitational potential at the surface of the Moon. Give a unit with your answer.

gravitational potential = ... unit ..... [3]

(b) An isolated uniform spherical planet has gravitational potential ϕ at its surface.

A particle of mass m is projected vertically upwards from the surface. The particle is given just enough kinetic energy to travel to an infinite distance away from the planet, escaping from the gravitational pull of the planet, without any additional work being done on it.

(i) Determine an expression, in terms of m and ϕ, for the gravitational potential energy E_P of the particle at the surface of the planet.

E_P = ... [2]

(ii) Show that the speed v at which the particle is projected upwards from the surface of the planet is given by

$$v = \sqrt{-2\phi}.$$

[3]

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(c) A particle is moving upwards at the surface of the Moon.

Use your answer in **(a)(ii)** and the expression in **(b)(ii)** to determine the minimum speed of this particle that will result in it escaping from the gravitational pull of the Moon.

speed = ... $m\,s^{-1}$ [2]

(d) Hydrogen may be assumed to be an ideal gas.
The mass of a hydrogen molecule is 3.34×10^{-27} kg.

Calculate the root-mean-square (r.m.s.) speed of a hydrogen molecule in hydrogen gas that is at a temperature of 400 K.

r.m.s. speed = ... $m\,s^{-1}$ [5]

(e) The surface of the Moon reaches temperatures of approximately 400 K when in direct sunlight.

Use your answers in **(c)** and **(d)** to suggest a reason why the Moon does not have an atmosphere consisting of hydrogen.

.....

..... [2]

[Total: 20]

Question 4 · Oscillations · 20 marks

Oscillating mass on a spring — energy lines, period & damping

3 An object is suspended from a vertical spring as shown in Fig. 3.1.

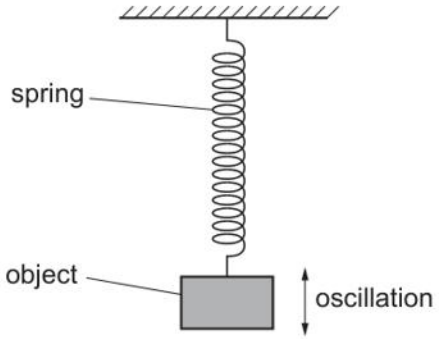


Fig. 3.1

The object is displaced vertically and then released so that it oscillates, undergoing simple harmonic motion.

Fig. 3.2 shows the variation with displacement x of the energy E of the oscillations.

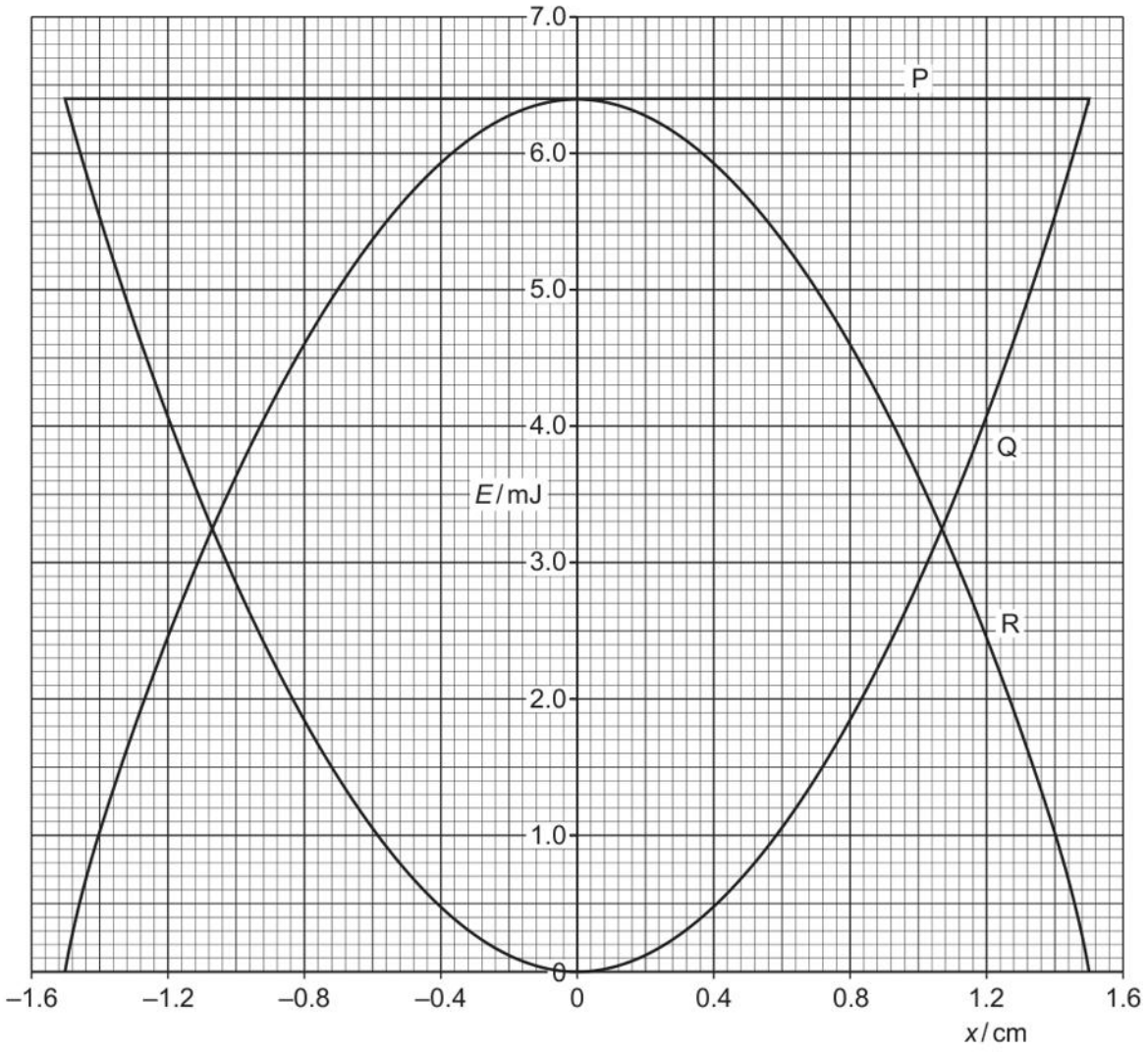


Fig. 3.2

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The kinetic energy, the potential energy and the total energy of the oscillations are each represented by one of the lines P, Q and R.

(a) State the energy that is represented by each of the lines P, Q and R.

P .....

Q .....

R ..... [4]

(b) The object has a mass of 130 g.
Determine the period of the oscillations.

period = ... s [8]

(c) **(i)** State the cause of damping.

.....

..... [2]

(ii) A light card is attached to the object. The object is displaced with the same initial amplitude and then released. During each complete oscillation the total energy of the system decreases by 8.0% of the total energy at the start of that oscillation.

Determine the decrease in total energy, in mJ, of the system by the end of the first 6 complete oscillations.

energy lost = ... mJ [4]

(iii) State, with a reason, the type of damping that the card introduces into the system.

.....

.....

..... [2]

[Total: 20]

Question 5 · Oscillations · 20 marks

Simple harmonic motion — defining equation, energy & damping

3 An object is suspended from a spring that is attached to a fixed point as shown in Fig. 3.1.

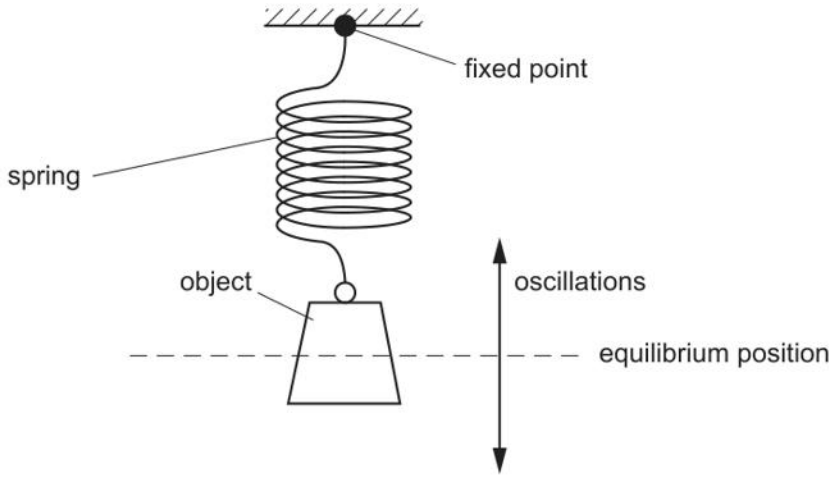


Fig. 3.1

The object oscillates vertically with simple harmonic motion about its equilibrium position.

(a) State the defining equation for simple harmonic motion. Identify the meaning of each of the symbols used to represent physical quantities.

.....

.....

..... [4]

(b) The variation with displacement x from the equilibrium position of the velocity v of the object is shown in Fig. 3.2.

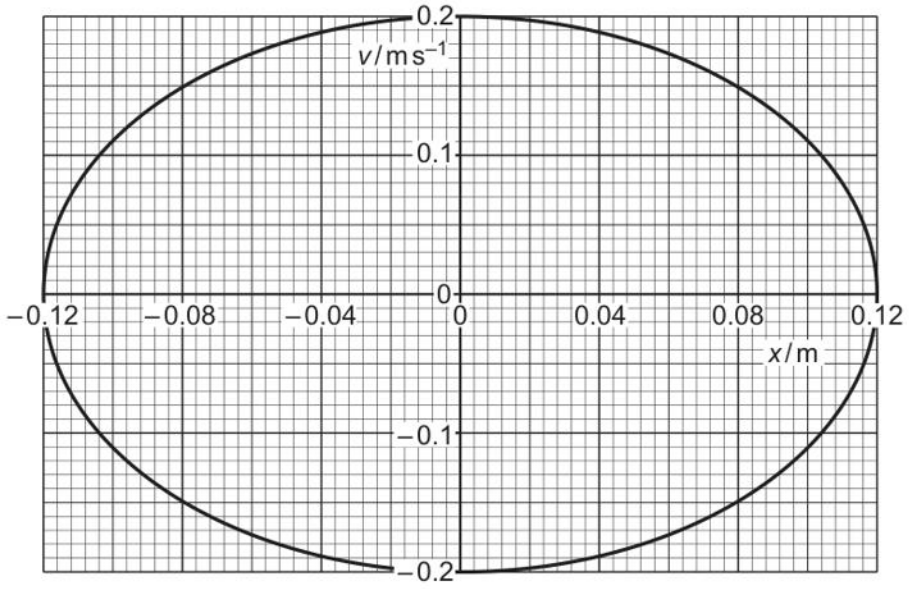


Fig. 3.2

The variation with x of the potential energy E_P of the oscillations of the object is shown in Fig. 3.3.

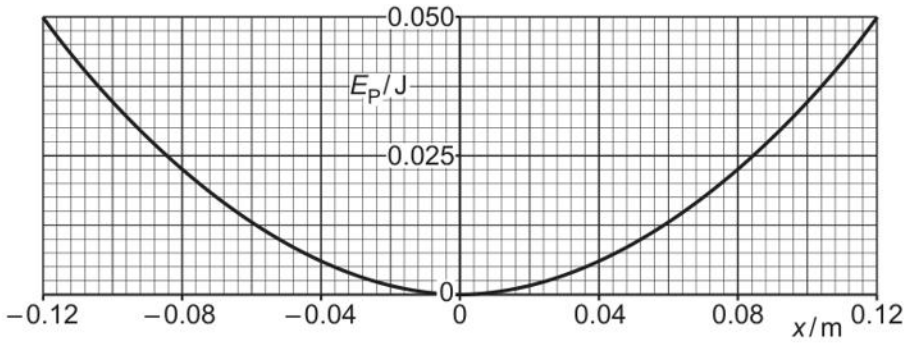


Fig. 3.3

Use Fig. 3.2 and Fig. 3.3 to:

(i) determine the amplitude x_0 of the oscillations

x_0 = ... m [2]

(ii) show that the angular frequency of the oscillations is 1.7 rad s^{-1}

[3]

(iii) determine the mass M of the object.

M = ... kg [3]

(c) The oscillations of the object are now lightly damped.

(i) State what is meant by damping.

.....

.....

..... [4]

(ii) Assume that the damping does not change the angular frequency of the oscillations.

On Fig. 3.2, sketch the variation with x of v when the amplitude of the oscillatio is 0.060 m. [4]

[Total: 20]